BLINKIT

Sales & Operations Dashboard

Project Documentation

Created by: Chanchal Khandkure

Tools: Microsoft Power BI • Power Query • DAX • Excel/CSV

Project focus	Sales analysis, product categories, outlet comparisons, ratings, and location-tier distribution.
Report structure	Three interactive pages: Overview, Product & Outlet Analysis, and Outlet Performance.
Design theme	Blinkit-inspired yellow, cream, olive, and charcoal palette.

1. Project Overview

This project presents an interactive Power BI dashboard for exploring grocery sales and outlet performance. It combines summary KPIs with product- and outlet-level visuals to support data exploration through slicers and cross-filtering.

Project objectives

- Summarize sales and rating metrics using KPI cards.
- Compare sales across outlet types, sizes, establishment years, and item categories.
- Explore the relationship between item visibility, sales, and ratings.
- Review sales distribution by location tier and fat-content group.
- Provide an outlet-level comparison using a performance table.

2. Tools and Technologies

Tool	Role in the project
Microsoft Power BI	Dashboard development and interactive visualizations.
Power Query	Data loading, review, and transformation workflow.
DAX	Measures and KPI calculations.
Excel / CSV	Dataset formats available in the project structure.

3. Dataset and Fields

The project folder includes a cleaned Excel dataset (BlinkIT-Grocery-Data_Clean.xlsx) and a raw CSV dataset (BlinkIT-Grocery-Data_Raw.csv). The report fields visible in the dashboard include:

- Item Type — product category
- Item Weight — product weight
- Item Visibility — product visibility value
- Fat Content — fat-content classification
- Sales — sales value
- Rating — rating field used in the report
- Outlet Identifier — outlet identifier
- Outlet Establishment Year — outlet establishment year
- Outlet Size, Outlet Type, and Outlet Location / Location Tier — outlet attributes

Use the exact column names in the Power BI model if they differ from these display labels.

4. Dashboard Pages and Visuals

Page 1 — Overview

Purpose: provide an executive summary of overall sales and outlet performance.

- KPI cards: Total Sales, Average Rating, Total Outlets, Total Items, and Total Categories.
- Sales trends by outlet establishment year.
- Sales by outlet type and item category.
- Top items by sales and average rating.
- Sales distribution by location tier.

Page 2 — Product & Outlet Analysis

Purpose: explore product attributes and sales differences across outlet characteristics.

- Sales versus item visibility (scatter chart).
- Sales by outlet size (column chart).
- Sales by item type (bar chart).
- Sales versus rating (scatter chart).
- Sales distribution by fat content (donut chart).
- Slicers: year/date, outlet type, location tier, outlet size, and fat content.

Page 3 — Outlet Performance

Purpose: compare outlets using sales, rating, visibility, and item-count metrics.

- Sales by outlet establishment year.
- Average rating by outlet type.
- Top outlets by total sales.
- Outlet performance summary table with average rating, average sales, total sales, total items, and average visibility.

5. Key Performance Indicators

KPI	Meaning	Validation note
Total Sales	Sales total in the current filter context.	Check consistent totals across visuals.
Average Rating	Average of the rating field.	Confirm the intended rating grain.
Total Outlets	Distinct outlet count.	Use distinct outlet identifiers.
Total Items	Item count defined by the report measure.	Confirm whether this means rows or unique items.
Total Categories	Distinct item-type count.	Check distinct category values.
Average Visibility / Weight	Average of the respective field.	Verify data types and blanks.

6. Example DAX Measures

These examples are templates based on fields visible in the report. Replace **BlinkIT** with the exact model table name and verify the column names before using them.

Total Sales

```
Total Sales = SUM ( BlinkIT[Sales] )
```

Average Rating

```
AVG RATING = AVERAGE ( BlinkIT[Rating] )
```

Total Outlets

```
TOTAL OUTLETS = DISTINCTCOUNT ( BlinkIT[Outlet Identifier] )
```

Total Categories

```
TOTAL CATEGORIES = DISTINCTCOUNT ( BlinkIT[Item Type] )
```

Average Visibility

```
AVG VISIBILITY = AVERAGE ( BlinkIT[Item Visibility] )
```

Average Item Weight

```
AVG ITEM WEIGHT = AVERAGE ( BlinkIT[Item Weight] )
```

Total Items depends on the intended definition. Use COUNTROWS(BlinkIT) only if the measure is meant to count rows; use DISTINCTCOUNT on the relevant product field if it should count unique products.

7. Visual Design and Color Palette

Use	Hex code	Recommended application
Blinkit yellow	#FCE006	Highlights, active navigation, selected accents.
Cream background	#F5F0D1	Report canvas.
Off-white	#FFFDF5	Visual and card backgrounds.
Charcoal	#29291F	Text, sidebar, table headers.
Olive	#77752B	Secondary accents and labels.
Muted gold	#B7AD64	Secondary chart series.
Border / gridline	#E8E2C9	Subtle boundaries and gridlines.

8. Project Folder Structure

The following structure reflects the files provided for the project:

Datasets/
■■■ BlinkIT-Grocery-Data_Clean.xlsx
■■■ BlinkIT-Grocery-Data_Raw.csv

Images/
■■■ Delivery.png
■■■ Home.png
■■■ Logo.jpg
■■■ Outlet.png
■■■ Profit.png
■■■ Store.png
■■■ Store_Shares.png

Screenshots/
■■■ Outlet Performance.png
■■■ Overview.png
■■■ Product & Outlet Analysis.png

Blinkit_Analysis.pbix
README.md

9. How to Open the Report

- Open Blinkit_Analysis.pbix in Microsoft Power BI Desktop.
- If prompted, update the source path to the dataset on your computer.
- Refresh the report and verify that visuals load correctly.
- Test each slicer and confirm that KPIs, charts, and tables respond as expected.
- Check all totals, titles, units, and measure definitions before sharing.

10. Quality Checks Before Publishing

- Confirm Total Sales is consistent across the report under the same filters.
- Verify that titles accurately describe each chart's axes and measure.
- Check that KPI cards show values and use consistent formatting.
- Ensure all three report pages are included in the PBIX file.
- Confirm screenshot filenames and README image links match exactly.
- Only publish dataset files if their license permits redistribution and they contain no sensitive information.

11. Project Outcome

This project demonstrates how a grocery sales dataset can be presented as an interactive Power BI report for exploring sales, product categories, ratings, and outlet performance. It showcases dashboard development, DAX calculations, interactive filtering, visual consistency, and business-oriented data storytelling.

Prepared by: Chanchal Khandkure

Portfolio focus: Data Analytics • Power BI • SQL • Python • Excel